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Experiment.1

Aim: To study and compare various wireless communication technologies and their characteristics.

Theory:

1. Bluetooth

- Range of Communication: Bluetooth typically has a range of about 10 to 100 meters, depending on the class of the device. Class 1 Bluetooth devices can achieve up to 100 meters, while Class 2 devices, which are more common in consumer products, usually have a range of 10-30 meters.
- Cost: Bluetooth modules and chips are generally inexpensive, with prices typically ranging from $1 to $5 for basic modules, though more sophisticated devices (like those supporting Bluetooth 5.0 or BLE) might cost more.
- Propagation Delay: The propagation delay for Bluetooth is quite low, typically around 1-2 milliseconds. This is because Bluetooth uses short-range radio frequencies, which results in quick communication with minimal latency.
- Power Consumption: Bluetooth Low Energy (BLE) is specifically designed for low power consumption. BLE devices can operate for months or even years on a small battery, making Bluetooth ideal for IoT devices and applications that require continuous but low-energy connectivity.
- Throughput: Classic Bluetooth supports data rates up to 3 Mbps, but Bluetooth Low Energy (BLE) supports a much lower data rate, typically up to 1 Mbps. The throughput is suitable for low-to-medium bandwidth applications, like transmitting small data packets for IoT sensors or peripherals.

2. Wi-Fi

- Range of Communication: Wi-Fi typically operates within a range of 30-100 meters indoors. The actual range depends on environmental factors such as walls, interference, and the frequency band used. Outdoors, the range can extend to several hundred meters if unobstructed.
- Cost: Wi-Fi modules and routers tend to be moderately priced, ranging from $5 for basic modules to $50 or more for advanced routers or high-performance modules that support newer standards like Wi-Fi 6.
- Propagation Delay: The propagation delay for Wi-Fi is relatively low, usually between 1 and 5 milliseconds. Modern Wi-Fi technologies (Wi-Fi 5 and Wi-Fi 6) have optimized

mechanisms to reduce latency, making Wi-Fi suitable for real-time applications like video conferencing and online gaming.

- Power Consumption: Wi-Fi generally has a moderate power consumption compared to Bluetooth or Zigbee. Devices that continuously transfer large amounts of data or stay connected to Wi-Fi networks will consume more power. Newer standards like Wi-Fi 6 have introduced power-saving features, but power consumption can still be significant for mobile devices.

3. Zigbee

- Range of Communication: Zigbee typically operates at ranges up to 100 meters in open space, but the actual range can vary based on obstacles and interference. In practical use, this range is often reduced in indoor environments.
- Cost: Zigbee modules are relatively inexpensive, typically costing between $2 to $10, depending on the manufacturer and features. It's a popular choice for cost-sensitive applications in home automation and IoT.
- Propagation Delay: Zigbee offers a low propagation delay, generally under 20 milliseconds. This makes it suitable for real-time monitoring and control in applications like home automation or industrial control systems.
- Throughput: Zigbee has a low throughput, typically around 250 kbps. This is suitable for applications that require small amounts of data to be transmitted infrequently, such as sensor networks or simple control devices.

4. LoRa (Long Range)

- Range of Communication: LoRa has one of the longest ranges among wireless communication technologies, reaching up to 15-20 kilometers in open space. In urban environments, the range may be reduced to a few kilometers depending on factors like interference and the presence of buildings.
- Cost: LoRa modules are affordable, generally costing between $10 to $30. This makes it an attractive option for large-scale IoT networks that require long-range communication without the need for cellular connectivity.
- Propagation Delay: The propagation delay for LoRa is higher compared to Bluetooth and Wi-Fi due to its long-range communication nature. Typical delays can be around 100 milliseconds, which is acceptable for many IoT applications, but not suitable for real-time applications that require low latency.
- Power Consumption: LoRa is optimized for low power consumption, allowing devices to run on small batteries for years. This is particularly useful for remote sensors and devices that need to operate over long periods without frequent recharging or battery changes.
- Throughput: LoRa supports a relatively low throughput, ranging from 0.3 kbps to 50 kbps. This is sufficient for transmitting small data packets, making it ideal for

applications like environmental monitoring, smart agriculture, and asset tracking where low data rates are needed.

5. 5G

- Range of Communication: 5G networks provide medium-range coverage, typically extending several kilometers from a cell tower. The range can vary based on the frequency band used—higher frequencies (mmWave) provide faster speeds but have a shorter range, while lower frequencies offer greater coverage but with lower speeds.
- Cost: 5G infrastructure is expensive due to the need for new cell towers, spectrum licenses, and advanced equipment. This can make the initial cost of 5G service higher compared to previous generations like 4G.
- Propagation Delay: 5G offers ultra-low latency, with round-trip times as low as 1 millisecond, making it ideal for applications that require near-instant communication, such as autonomous vehicles, augmented reality, and industrial automation.
- Power Consumption: Power consumption is generally higher for 5G compared to older technologies like 4G, especially for devices using higher frequency bands (mmWave). However, the power consumption is expected to decrease as 5G technology evolves and power-saving features are implemented.
- Throughput: 5G provides extremely high throughput, with speeds ranging from 100 Mbps to 10 Gbps in ideal conditions. This makes it suitable for applications requiring very high data rates, such as high-definition video streaming, cloud gaming, and real-time data analytics.

6. NFC

- NFC stands for Near Field Communication, a technology that allows devices to communicate wirelessly over short distances. NFC is used in many devices, including smartphones, tablets, and consumer electronics.
- NFC uses short-range wireless signals to connect devices that are within a few centimeters of each other. NFC is a peer-to-peer technology that automatically connects compatible devices.
- NFC can be used to exchange data, make payments, and connect devices.

7. Infrared (IR)

- Range of Communication: Infrared communication typically operates at very short ranges, usually between 1 and 5 meters, depending on the power of the infrared transmitter and receiver.
- Cost: IR devices are among the least expensive, with typical costs for modules being under $1.

- Propagation Delay: The propagation delay for infrared communication is minimal, typically around 1 millisecond, due to the short distance the signal travels.
- Power Consumption: IR communication is energy-efficient, especially in low-power modes, as it is used for short-range, low-data applications.
- Throughput: The throughput for infrared is low, generally around 1 Mbps, which limits its use to applications like remote control devices, short-range data transmission, and simple signaling.

8. Satellite Communication

- Range of Communication: Satellite communication provides global coverage, as satellites in geostationary orbit can cover large areas of the Earth's surface. Communication is possible anywhere there's a satellite signal, including remote and rural locations.
- Cost: Satellite communication is expensive, both in terms of infrastructure (satellite launches, ground stations) and service fees. The cost is typically much higher than other communication technologies.
- Propagation Delay: Propagation delay is significant, typically around 250 to 500 milliseconds, because the signals must travel to and from satellites in geostationary orbit, which is about 35,786 kilometers above Earth.
- Power Consumption: The power consumption for satellite communication is high, as it requires large antennas and powerful transmitters to send and receive signals over long distances.
- Throughput: Satellite communication throughput can vary widely depending on the system used. Traditional satellite systems may offer low throughput (kbps), while newer systems like SpaceX's Starlink can offer much higher speeds (up to several Gbps).

9. Cellular Communication (4G/5G)

- Range of Communication: Cellular networks, including 4G and 5G, typically cover ranges of a few kilometers, depending on network density and frequency band. Lower-frequency bands provide wider coverage, while higher-frequency bands (used in 5G) provide faster speeds but shorter ranges.
- Cost: Cellular infrastructure (towers, spectrum licenses) is expensive, which translates to higher service costs. However, devices and modules are more affordable than satellite communication.
- Propagation Delay: Propagation delays are moderate, with 4G typically having latencies of around 20-50 milliseconds, and 5G offering significantly lower latency, potentially as low as 1-10 milliseconds.
- Power Consumption: Power consumption is moderate for cellular devices, but 5G can be more power-hungry than 4G, especially when using high-frequency bands. However, power efficiency is improving as 5G evolves.

Summary Table:

Technology	Range of Communication	Cost	Propagation Delay	Power Consumption	Throughput
Bluetooth	10-100 meters	₹100-500	1-2 ms	Low	1-3 Mbps
Wi-Fi	30-100 meters (indoor)	₹500-₹2500	1-5 ms	Moderate	1-10 Gbps
Zigbee	100 meters	₹200 - 1500	<20 ms	Very Low	250 kbps
LoRa	15-20 km	₹500-₹4000	100 ms	Very Low	0.3-50 kbps
5G	Several kilometers	15000 - 25000	1 ms	Moderate-High	100 Mbps - 10 Gbps
NFC	10 cm	₹50-₹500	Low	Very Low	106 to 424 kbit/s
Infrared	1-5 meters	₹50-₹100	1 ms	Low	1 Mbps
Satellite	Global	₹50000 - ₹100000	250-500 ms	High	Varies (kbps to Gbps)
Cellular (4G/5G)	Few kilometers	₹500-₹2000	20-50 ms (4G), 1-10 ms (5G)	Moderate	1 Gbps (4G), 10 Gbps (5G)

Conclusion:

Wireless communication technologies have evolved significantly, each offering unique characteristics in terms of range, speed, and application. Understanding these differences helps in selecting the most suitable technology for specific use cases.